Cardiovascular Examination — OSCE Checklist

Deakin MD2 · CR3 · Time: 8 min (+2 min reading)

Tick each step as you perform or verbalise it. **Shaded rows (*)** *are optional, verbalised or done if indicated.* † *= added from standard teaching (Talley & O'Connor), not in the Deakin guideline. Order:* **General → Vitals → Hands → Arms (pulse, BP) → Face → Neck (carotid, JVP) → Praecordium → Back → Abdomen → Legs → Other**. *Patient at* **45°**. *Expose the chest only when needed.*

1. Introduction

☐	Hand hygiene; introduce self; confirm patient ID
☐	Explain the exam and gain consent: upper body clothing off, exposed only when needed
☐	Offer a chaperone; ensure privacy, comfort and warmth
☐	Position: on the bed at **45°**, supported with pillows
☐	Equipment: stethoscope, sphygmomanometer, torch, ruler †

2. General inspection and vital signs

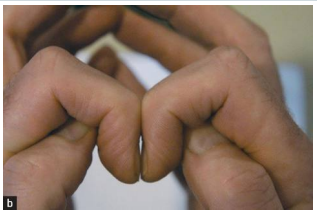
	Step	What to do / say	Abnormal
☐	Well or unwell?	Pallor, jaundice, cachexia, sweaty/diaphoretic	Shock, heart failure, endocarditis
☐	Breathing	Respiratory effort, rate, distress	Pulmonary oedema
☐	Body features	Marfan's, Turner's, trisomy 21	Associated cardiac lesions
☐	Mobility, expression	Walked in? How? Anxious or in pain?	
☐	Bedside clues	Medications, glucometer/CGM, oxygen, IV cannula/fluids, 'fluid restriction' or 'low salt' sign, cigarettes/vapes	
☐	Vital signs	HR, BP, RR, temp, SpO_2, BGL	

3. Hands and arms

	Step	What to do / say	Abnormal
☐	Clubbing	Schamroth's window (nail-fold angle)	Infective endocarditis, cyanotic heart disease
☐	Splinter haemorrhages	Nail beds	Infective endocarditis
☐	Osler's nodes	Red, raised, **painful** 0.5–1 cm nodules on finger/toe pulps, thenar/hypothenar	Infective endocarditis
☐	Janeway lesions	**Non-tender** red lesions on palms or finger pulps	Infective endocarditis
☐	Colour	Peripheral cyanosis; nicotine staining; pallor of palmar creases	Low output; smoking; anaemia
☐	Temperature	Warm (vasodilated), sweaty, tremor; or cool, peripherally shut down	Sepsis, thyrotoxicosis; shock
☐	Tendon xanthoma	Yellow deposits over extensor tendons	Hypercholesterolaemia
☐	Radial pulse	Rate and rhythm	Tachy-/bradycardia; irregularly irregular = AF
☐	Radio-radial delay	Both radials together: timing and volume	Aortic dissection, subclavian stenosis
☐	Radio-femoral delay	Radial + femoral together	Coarctation of the aorta
☐	Arms	IV drug injection scars	Endocarditis risk
☐	Blood pressure	Manual BP, **both arms**, **lying and standing**	Hypertension; postural drop



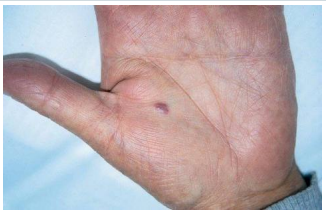
Schamroth's sign — clubbed (no window)



Normal — diamond window



Splinter haemorrhages

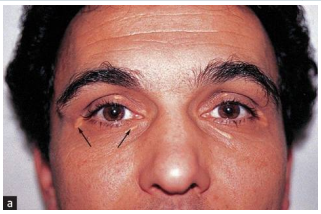


Janeway lesion

4. Face and neck

	Step	What to do / say	Abnormal
☐	Face	Expression (pain, apprehension); mitral facies (malar flush)	Mitral stenosis
☐	Eyes	Xanthelasmata, corneal arcus; jaundice; conjunctival pallor †	Hyperlipidaemia; congestive hepatopathy; anaemia
☐	Mouth	Central cyanosis; high-arched palate; poor dentition	Hypoxaemia; Marfan's; endocarditis source
☐	Carotid pulse	**One side at a time**: fingers behind neck, thumb at/below cricoid; volume and character	Slow-rising (AS), collapsing (AR)
☐	Carotid bruit	Bell; patient holds breath; each side; tell them to breathe again	Carotid stenosis
☐	JVP — position	45°, head tilted up and turned slightly away; look at **internal** jugular (between heads of SCM)	
☐	JVP — see it	Torch at an oblique angle; look for the **double flicker**	

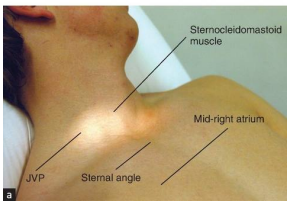
	Step	What to do / say	Abnormal
☐	JVP — measure	Ruler held horizontally from top of pulsation; measure vertical height above sternal angle	>3 cm = raised (right heart failure, fluid overload)
☐	JVP — character	Waveform	Large v waves (TR), absent a waves (AF) †



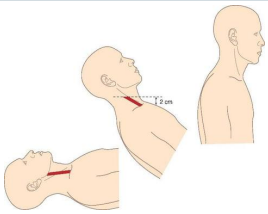
Xanthelasmata



Corneal arcus



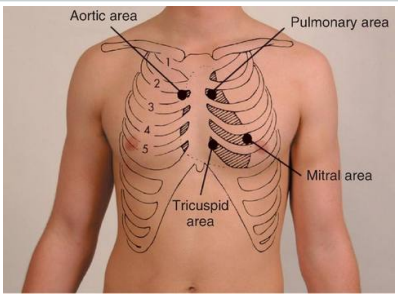
JVP and sternal angle



JVP height changes with position

5. Praecordium

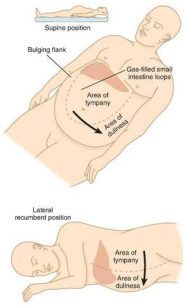
	Step	What to do / say	Abnormal
Inspect			
☐	Scars	Median sternotomy, thoracotomy †	Previous cardiac surgery
☐	Chest wall	Pectus excavatum, other deformity	
☐	Devices	Pacemaker or defibrillator box	
☐	Pulsations	Visible apex beat	
Palpate			
☐	Apex beat	Supine or left lateral: normally **5th ICS, mid-clavicular line**; presence, position, character	Displaced = cardiomegaly; heaving = pressure-loaded; diffuse = volume-loaded
☐	Absent apex?	Consider thick chest wall, emphysema, pericardial effusion, dextrocardia	
☐	Parasternal impulse	Heel of hand at left sternal edge	Heave = right ventricular enlargement
☐	Thrills	Flat hand over apex, left sternal edge, then base (aortic and pulmonary areas)	Palpable murmur; time with apex beat
Auscultate (time with carotid pulse †)			
☐	Mitral area	Apex: 5th left ICS, mid-clavicular line	S1, S2; murmurs
☐	Tricuspid area	5th left ICS, at the left sternal edge	
☐	Pulmonary area	2nd left ICS	
☐	Aortic area	2nd right ICS	
☐	Heart sounds	S1 (mitral + tricuspid closure), S2 (aortic + pulmonary); added sounds?	Murmur: systolic or diastolic, grade (Levine)
☐	Manoeuvres*	Left lateral + bell at apex; sit forward in full expiration at left sternal edge †	Mitral stenosis; aortic regurgitation



Auscultation areas

6. Back and abdomen

	Step	What to do / say	Abnormal
☐	Back	Sit forward: inspect for scars and deformity	Previous surgery
☐	Lung bases	Percuss and auscultate	Crackles, effusion (heart failure)
☐	Sacral oedema	Press over sacrum	Fluid overload
☐	Liver	Start in RIF, hand parallel to costal margin, advance on each breath out; span (normal <13 cm)	Enlarged, tender (congestion); pulsatile (TR †)
☐	Spleen	Lower mid-abdomen towards left costal margin; roll patient towards you	Splenomegaly (endocarditis)
☐	Ascites	Percuss midline → flanks; if dull, test shifting dullness (roll, wait 30–60 s)	Right heart failure
☐	Aorta	Palpate left of midline	Expansile = AAA
☐	Femoral arteries	Palpate and auscultate (midway pubic symphysis–ASIS)	Weak pulse, bruit (PVD)



Shifting dullness



Peripheral pulses

7. Legs

	Step	What to do / say	Abnormal
☐	Inspect	Scars (vein harvest †), ulcers, hair loss, shiny skin, trophic nails, colour (white, cyanosed, red), clubbing of toes	Peripheral vascular disease
☐	Oedema	Ankle swelling; press for pitting; note the upper level	Heart failure
☐	Calves	Tenderness, swelling †	DVT
☐	Popliteal pulse	Centre of popliteal fossa, knee slightly bent	
☐	Posterior tibial	Behind medial malleolus	Absent = PVD
☐	Dorsalis pedis	Lateral to extensor hallucis longus tendon	Absent = PVD
☐	Capillary return	Press toe, watch colour return	Slow = poor perfusion
☐	Buerger's test*	If slow: raise legs until pale, then hang over the side of the bed	Delayed colour, cyanosis = arterial insufficiency

8. Closing and presenting

☐	Completion: "I would also do urinalysis and fundoscopy, and review the ECG and observation chart †."
☐	Simple explanation of findings; anything you can't answer → "I'll pass that on to someone senior"
☐	Ask about concerns/questions; check understanding and comfort; patient can redress
☐	Hand hygiene; return equipment; document findings
☐	Present: "Mr X is comfortable at rest; pulse 72 regular, BP 125/80. No peripheral stigmata of endocarditis or hyperlipidaemia. JVP not raised. Apex beat 5th ICS mid-clavicular line, undisplaced. Heart sounds dual with no added sounds. Lung bases clear, no peripheral oedema, peripheral pulses present."

9. Quick reference

Category	Systolic	Diastolic
Optimal	<120 and	<80
Normal	120–129 and/or	80–84
High normal	130–139 and/or	85–89
Grade 1 HTN	140–159 and/or	90–99
Grade 2 HTN	160–179 and/or	100–109
Grade 3 HTN	≥180 and/or	≥110
Isolated systolic	≥140 and	<90

Grade	Levine murmur grading
1/6	Very soft, not heard at first
2/6	Soft, heard almost immediately
3/6	Moderate, no thrill
4/6	Loud, thrill just palpable
5/6	Very loud, thrill easily palpable
6/6	Heard without stethoscope on chest

Murmur †	Timing	Best heard	Key signs
Aortic stenosis	Ejection systolic	Aortic area → carotids	Slow-rising pulse, narrow pulse pressure
Mitral regurgitation	Pansystolic	Apex → axilla	Displaced apex, AF
Aortic regurgitation	Early diastolic	Left sternal edge, sitting forward, expiration	Collapsing pulse, wide pulse pressure
Mitral stenosis	Mid-diastolic rumble	Apex, left lateral, bell	Malar flush, AF, tapping apex

- **Heart failure †**: tachycardia, raised JVP, displaced apex, S3, bibasal crackles, pleural effusion, tender liver, sacral and ankle oedema.
- **Infective endocarditis**: clubbing, splinters, Osler's nodes, Janeway lesions, new murmur, splenomegaly, fever.
- BP: one reading cannot diagnose hypertension (classification for reference only).